MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) REGULAR EXAMINATION

March 2026 Fourth SEMESTER

QP Code 25901090

Discipline Specific Core Course (DSC) - **MG4DSCBCH202 - METABOLISM OF CARBOHYDRATES, PROTEINS AND LIPIDS** (2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice questions

Answer all questions Each question carries 1 marks

[1x5 = 5]

1. The only source of energy in erythrocytes is
   Ans-Option d- Glycolysis [U] / [CO1]
2. Select the product formed from the ketogenic amino acids:
   Ans- Option c- Acetyl CoA [U] / [CO3]
3. What is the role of acyl CoA dehydrogenase enzyme in oxidation of palmitate?
   Ans Option a- Oxidation [U] / [CO4]
4. Which activity of the debranching enzyme explains the transfer of glucose residues from a branch?
   Ans Option a-Amylo-α-1,6-glucosidase
   [K] / [CO2]
5. Select the amino acid that undergoes non - oxidative deamination by the action of dehydratase:
   Ans Option a- Serine [E] / [CO3]

Part B

Fill in the Blanks questions

Answer all questions Each question carries 2 marks

[2x5 = 10]

6. The synthesis of ATP from ADP and Pi is driven by ETC in a process called_oxidative phosphorylation_____ [U] / [CO2]
7. Inhibition of SGLT-1 will mainly affect absorption of _glucose____ and _sodium_ [U] / [CO1]
8. Condensation reaction forms _acetoacetyl ACP / beta ketoacyl ACP [K] / [CO4]
9. The enzyme deficient in albinism is tyrosinase__ [U] / [CO3]
10. The enzyme succinate thiokinase is also known as ___succinyl coAsynthase ___ [K] / [CO2]

Part C

Short Answer Type Questions

Answer any 5 questions Each question carries 3 marks

[3x5 = 15]

11. What is glycogenesis? When is it activated?
    Glycogenesis is the process of glycogen synthesis, in which glucose molecules are added to chains of glycogen for storage. This process is activated during rest periods and also activated by insulin in response to high glucose levels, for example after a carbohydrate-containing meal. [K] / [CO2]
12. Evaluate protein digestion and assess the role of brush-border peptidases in the final stages of protein digestion.
    Protein digestion is the process by which dietary proteins are broken down into amino acids so that they can be absorbed and used by the body. This process begins in the stomach, where the acidic environment (hydrochloric acid) denatures proteins and activates the enzyme pepsin from its inactive form, pepsinogen. Pepsin breaks proteins into smaller polypeptides.

    In the small intestine, protein digestion continues with the help of pancreatic enzymes such as trypsin, chymotrypsin, elastase, and carboxypeptidases, which further break polypeptides into smaller peptides and amino acids.
    intestinal peptidases present on the brush border of intestinal cells convert peptides into free amino acids, dipeptides, and tripeptides. [E] / [CO1]
13. List the major pathways of lipid metabolism
    (Mention any three of the following)

The major pathways of lipid metabolism include:

1. Digestion & absorption of dietary lipids
2. Transport of lipids in blood as lipoproteins
3. Storage of lipids mainly in adipose tissue
4. Mobilization of stored fat (lipolysis)
5. Oxidation of fatty acids (mainly β-oxidation) to produce energy
6. Synthesis of fatty acids & triglycerides
7. Cholesterol metabolism including its synthesis, utilization and excretion.

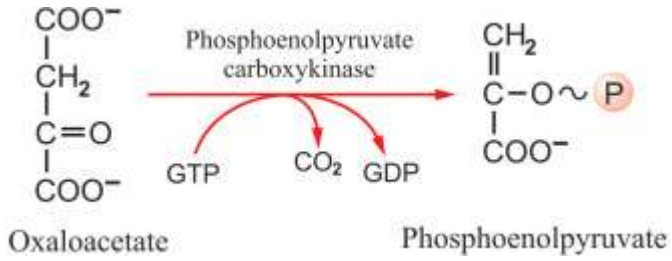
[K] / [CO1]

14. Explain and illustrate why urine turns dark on standing in alkaptonuria.
    Alkaptonuria is a rare inborn error of metabolism caused by a deficiency of the enzyme homogentisic acid oxidase, which is involved in the metabolism of the amino acids phenylalanine and tyrosine. Due to this enzyme defect, homogentisic acid accumulates in the body and is excreted in the urine.

    A characteristic feature of alkaptonuria is that the urine turns dark brown or black on standing due to oxidation of homogentisic acid. [U] / [CO3]
15. Why is α-oxidation necessary in fatty acid metabolism?
    α-Oxidation occurs in the peroxisomes and is important for the oxidation of branched-chain fatty acids, such as phytanic acid. It involves the removal of one carbon atom at a time and is necessary because branched fatty acids cannot undergo normal β-oxidation directly.[U] / [CO4]

16. Demonstrate the action of PEP carboxylase in gluconeogenesis

In the cytosol Phosphoenol pyruvate carboxykinase converts oxaloacetate to PEP by losing a molecule of CO2 and adding a high energy phosphate. GTP is used in this reaction.



[U] / [CO2]

17. Explain the role of oxidative deamination in the removal of amino groups from amino acids.

In oxidative deamination, liberation of free ammonia from amino acids is coupled with oxidation.

Oxidative deamination

It is catalyzed by one of the following enzymes:

1. L-amino acid oxidases
2. D-amino acid oxidases
3. Glutamate dehydrogenase [U] / [CO3

Part D

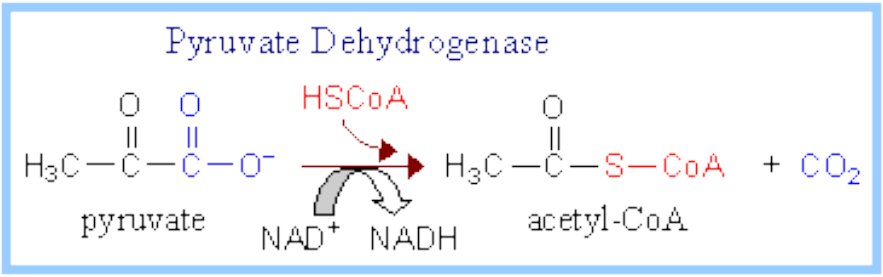
Short Essay Type Questions

Answer any 4 questions Each question carries 5 marks

[5x4 = 20]

18. Explain the reaction catalyzed by pyruvate dehydrogenase complex.

In the cytosol Phosphoenol pyruvate carboxykinase converts oxaloacetate toThe Pyruvate derived from glycolysis is oxidatively decarboxylated to acetyl-CoA by the pyruvate dehydrogenase (PDH).This is the link between TCA cycle and hydrolysis. The PDH reaction occurs in mitochondria.Pyruvate with the help of a carrier can enter the mitochondria from the cytoplasm. Pyruvate Dehydrogenase complex catalyzes oxidative decarboxylation of pyruvate, to form acetyl-CoA. The overall reaction is



Pyruvate Dehydrogenase Complex found only in mitochondria is composed of three enzymesand five coenzymes

1. Pyruvate dehydrogeanse (E1)
2. Dihydrolipoamide transacetylase (E2)
3. Dihydrolipoamide dehydrogenase (E3)
1. Thiamine Pyrophosphate (TPP)
2. Lipoic Acid - lipoamide
3. FAD
4. NAD^+
5. CoASH [U] / [CO2]

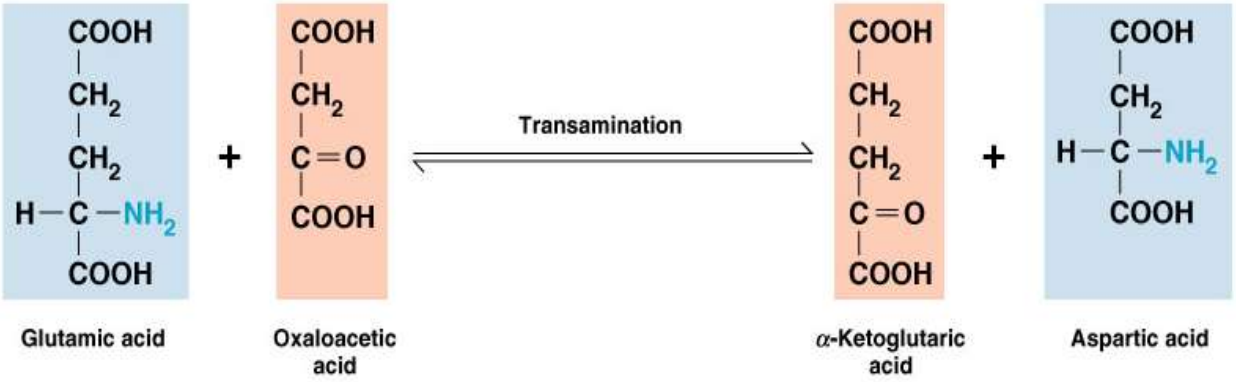
19. Appraise the net ATP yield during the conversion of 2 molecule of pyruvate to acetyl CoA.

Pyruvate dehyrogenase reaction 2 Pyruvate ⟶ 2 acetyl-CoA	2 NADH	6 ATP

[U] / [CO2]

20. Describe how transamination reactions contribute to the formation of glutamate.

Transaminations are freely reversible reactions catalyzed by transaminases (aminotransferases). Amino group of α-amino acid is exchanged with oxo group of 2-oxoacidduring transamination. The amino group is transferred by cofactor pyridoxal phosphate (PLP, derivative of vitamin B6) to oxoacid.Most of the amino acids undergo transamination in their degradation. Enzymes aspartate aminotransferase (AST) and alanine aminotransferase (ALT) are examples of transaminases that are normally detected as markers of potential damage to the liver cells.



[U] / [CO3]

21. Explain why the urea cycle is considered an energy-consuming pathway.
    For attempt give full marks [U] / [CO3]
22. Describe the digestion of dietary triglycerides in gastrointestinal tract.

Digestion of lipids is the process by which dietary fats are broken down into simpler molecules so they can be absorbed by the body. Although lipid digestion begins in the mouth and stomach, the small intestine is the main site where most fat digestion occurs.

1.Mouth:

Digestion starts with lingual lipase, secreted by salivary glands. It acts mainly on short- and medium-chain triglycerides. Its role is limited in adults but is important in infants.

2.Stomach:

In the stomach, gastric lipase continues the digestion of triglycerides by breaking them into diglycerides and free fatty acids. Churning of food helps mix fat with enzymes, but only a small amount of fat digestion occurs here.

3.Small Intestine (Major Site):

The **small intestine** is the principal site for lipid digestion. When partially digested fat enters the duodenum, it stimulates the release of bile and pancreatic juice, which are essential for effective fat digestion.

Role of Bile:

Bile is produced by the liver and stored in the gallbladder. It contains bile salts that **emulsify fats**, breaking large fat globules into tiny droplets. Emulsification increases the surface area of fats, making them more accessible to digestive enzymes. Bile salts also help in the formation of micelles later during absorption.

Pancreatic Enzymes:

The pancreas secretes several enzymes into the small intestine:

Pancreatic lipase is the most important enzyme and converts triglycerides into **2-monoglycerides and free fatty acids**. Its activity is aided by **colipase**.

Phospholipase A_2 digests phospholipids into lysophospholipids and fatty acids.

Cholesterol esterasehydrolyzes cholesterol esters into free cholesterol and fatty acids.

End Products:

The digestion of lipids in the small intestine produces **free fatty acids, monoglycerides, cholesterol, and lysophospholipids**, which are then ready to be absorbed by the intestinal mucosal cells. [U] / [CO3]

23. Detail the stages of fatty acid oxidation.

Activation of fatty acids in the cytosol (1 mark)

Transport of fatty acids into mitochondria (Carnitine shuttle) (1.5 marks)

β-oxidation (2.5 marks)

[U] / [CO4]